

Shiksha Setu: A Measurement-Driven, Local-First AI Tutoring Platform for Multilingual Indian Education

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Abstract—Artificial intelligence holds transformative potential for education in linguistically diverse nations, yet existing systems overwhelmingly depend on cloud infrastructure, operate primarily in English, and lack alignment with national curricula. This paper presents Shiksha Setu, a local-first AI tutoring platform whose ingestion pipeline addresses the complete National Council of Educational Research and Training (NCERT) catalog—559 textbooks spanning classes 1 through 12 in three publication languages—and which delivers retrieval-augmented instruction in eleven Indian languages. The curriculum, its embeddings and the whole of retrieval are local: no textbook content, index or search traffic leaves the machine. Generation is a hosted call, so a question’s text does leave it, and the paper is explicit about which half is which rather than claiming the system is offline. All results reported here are measured against an ingested corpus of 153 textbooks spanning all twelve classes (42,995 indexed passages); the remaining books of the taught curriculum are unobtainable rather than pending, chiefly because NCERT publishes no archive for them. The system employs BGE-M3 multilingual embeddings with HNSW-indexed vector search over PostgreSQL, retrieving an on-topic passage at rank 1 for 16 ± 1 of 22 queries spanning eleven Indian languages against the English-medium corpus, measured against four alternative configurations: a lexical baseline retrieves nothing at all, dense retrieval alone 11, and query rewriting alone 14, so the deployed pooling of both arms is justified by measurement rather than by reasoning. The residual failures are traced to a compound-term effect that persists unchanged when the corpus is enlarged fivefold. Through half-precision inference, the embedding pipeline reduces its memory footprint from 2,166 MB to 1,083 MB at a measured cosine fidelity of 0.999999 against full precision, enabling deployment on machines with as little as 4 GB of RAM—verified through kernel-level peak resident set size measurement rather than arithmetic estimation. The ingestion pipeline processes the obtainable corpus at 140 seconds per book—a $1.38\times$ speedup over sequential execution, established by re-ingesting an identical book set with prefetching disabled—and the platform discovers and addresses two classes of corpus corruption: legacy Devanagari fonts that render 66% of Hindi textbooks as Latin gibberish, and mathematical symbol erasure in PDF text layers. A security audit identifies and repairs twelve vulnerabilities, including a safety pipeline that silently failed open and a JWT implementation that accepted refresh tokens as access credentials. Reading support for dyslexic learners introduces akshara-level segmentation for nine Brahmic scripts, replacing the Latin-alphabet assumptions of conventional readability tooling. The system is validated by 445 automated tests with zero failures across unit, integration, and end-to-end suites.

Index Terms—Retrieval-Augmented Generation, Multilingual Education, Local-First AI, Indian Languages, Curriculum-

Aligned Tutoring, Accessibility, NCERT

I. INTRODUCTION

India is home to 1.4 billion people speaking 22 constitutionally recognized languages, written in at least 13 distinct scripts. The National Council of Educational Research and Training (NCERT) publishes the curriculum that governs instruction for over 250 million school-age children, yet its textbooks exist only in English, Hindi, and Urdu. A Tamil-speaking student in rural Tamil Nadu, a Marathi-speaking student in Maharashtra, and a Bengali-speaking student in West Bengal all study the same curriculum—but none of them can read it in their mother tongue unless a human teacher translates it in real time.

Artificial intelligence can bridge this gap, but the dominant paradigm of cloud-hosted large language models introduces three structural failures for Indian education:

Language exclusion. The overwhelming majority of educational AI systems operate in English as their primary language. For the estimated 800 million Indians whose primary language is not English [1], this renders such systems inaccessible at the point of need.

Infrastructure dependence. Cloud-based solutions require consistent high-bandwidth internet connectivity and impose per-query costs through API billing. In Tier 2 and Tier 3 Indian cities and rural areas, this dependency makes sustained educational use economically impractical.

Pedagogical misalignment. General-purpose language models lack awareness of curriculum structure, grade-appropriate complexity, and the specific conceptual progression of NCERT instruction. A class 6 student asking about chemical reactions requires a fundamentally different explanation from a class 12 student on the same topic.

This paper presents Shiksha Setu, a platform that addresses the first and third failures and *partially* the second, through a measurement-driven engineering approach.

The qualification is deliberate. Retrieval is entirely local—the corpus, its embeddings and the vector index never leave the machine—but generation is served by a hosted endpoint, so a question’s text does leave it and does incur a per-query cost. The system therefore removes the dependency for the component that dominates storage and latency without removing it altogether. Local generation is configured and

available (Section III-B), but a language model cannot share a 4 GB machine with the embedder, so the offline configuration and the 4 GB configuration are alternatives rather than the same system. We state this plainly because “local-first” is otherwise read as “offline”, and the measurements in this paper were taken in the hosted configuration. Every claim is backed by an experiment whose procedure, data, and result are reported—including experiments that produced wrong results before being corrected. The contributions are:

- 1) An ingestion pipeline addressing all 559 NCERT textbooks, with automatic detection and rejection of legacy-font corruption that renders 27 of 41 Hindi books unreadable—discovered only through measurement.
- 2) A cross-lingual retrieval system enabling questions in eleven Indian languages to retrieve relevant passages from an English corpus, with a controlled ablation over 22 queries in eleven languages establishing that query rewriting and direct embedding fail on disjoint inputs, so that pooling them retrieves correctly for 16 ± 1 of 22 where either alone manages 11 or 14, and a lexical baseline none.
- 3) A memory optimization strategy verified through kernel-level measurement enabling the serving pipeline to operate within a 4 GB memory budget, correcting an earlier arithmetic estimate that understated consumption by 31%.
- 4) Akshara-level readability measurement for nine Brahmic scripts, replacing Latin-alphabet syllable counting with script-appropriate segmentation.
- 5) A security audit documenting twelve vulnerabilities—all repaired—including a safety pipeline that failed open due to an import error.
- 6) A grounding metric detecting when fluent, confident answers have drifted from source material—a failure mode indistinguishable from a correct answer without quantitative measurement—calibrated on 256 answer/passage pairs rather than on hand-checked examples, and compared against three alternative estimators.

II. RELATED WORK

A. Retrieval-Augmented Generation

Lewis et al. [2] introduced retrieval-augmented generation (RAG) for knowledge-intensive NLP tasks, establishing the paradigm of conditioning language model output on retrieved passages. Shiksha Setu adopts this architecture with a critical modification: a grade-level filter inserted between retrieval and generation, because the same topic at class 6 and class 12 requires different explanations. Karpukhin et al. [3] demonstrated the superiority of dense passage retrieval over BM25. Ma et al. [4] showed that query rewriting before embedding improves retrieval quality—the technique that resolves the romanized Hindi failure of Section V-B. Nogueira and Cho [5] established passage re-ranking with cross-encoder models; the BGE-Reranker component exists in the codebase but is not yet wired into the retrieval path.

B. Multilingual Embeddings and Indian Language NLP

Chen et al. [6] presented BGE-M3, a multilingual embedding model supporting over 100 languages in a unified 1024-dimensional vector space. Its cross-lingual property—semantically equivalent passages in different languages map to nearby vectors—allows a Hindi question to retrieve an English passage without explicit translation. Gala et al. [7] developed IndicTrans2 covering all 22 scheduled Indian languages, serving as the translation backbone. Kakwani et al. [1] provided monolingual corpora and evaluation benchmarks for Indian languages. Khanuja et al. [8] trained MuRIL on transliterated text, directly relevant to the romanized Hindi failure documented in Section V-B.

C. Output Language Drift and tRAG

Recent literature categorizes multilingual retrieval into frameworks such as MultiRAG (direct retrieval) and tRAG (translation-RAG). The platform described in this work formally implements a tRAG architecture [9], rewriting queries to a high-resource language (English) before retrieval. This architectural choice unknowingly resolves a newly documented vulnerability in large language models: Output Language Drift [10]. Studies show that when a multilingual RAG system processes a query in a target language but retrieves context in a dominant high-resource language, the model mixes languages and outputs the final answer in the high-resource language. The strict separation of the generation language from the delivery language via the translation layer elegantly neutralizes this cognitive drift.

D. Vector Search

The retrieval layer uses the HNSW graph structure proposed by Malkov and Yashunin [11], implemented through the pgvector extension for PostgreSQL. The index parameters ($m = 16$, $ef_construction = 64$) balance build time against recall; measured search latency of approximately 20 ms for 12-nearest-neighbor queries confirms sub-interactive response times. Reimers and Gurevych [12] established the sentence-level embedding formulation used to load BGE-M3.

E. Reading Accessibility and Brahmic Scripts

Zorzi et al. [13] found that extra-large letter spacing improved reading speed and halved errors in dyslexic children—the single typographic intervention with strong experimental support and the default accessibility control in the platform. Rello and Baeza-Yates [14] and Kuster et al. [15] found no reliable advantage for dyslexia-specific typefaces. Nag [16] and Nag and Snowling [17] demonstrated that akshara-based literacy develops differently from alphabetic literacy, motivating the akshara-level segmentation described in Section VII.

F. Efficiency and Readability

Kincaid et al. [18] derived the Flesch-Kincaid grade-level formula defined on English syllable counts—which is why the platform refuses to compute it for Devanagari text. Xu

et al. [19] demonstrated that generic simplification routinely loses content, motivating the measure-and-keep-the-easier approach. Micikevicius et al. [20] established the safety of half-precision arithmetic for neural network inference, providing the theoretical basis for the fp16 optimization that halves the BGE-M3 memory footprint.

G. Comparison with Existing Systems

Table I compares this work with the systems an Indian student is most likely to encounter: Khanmigo [21], Khan Academy’s GPT-4 tutor; BYJU’S, the largest Indian ed-tech platform; and Bhashini [22], the Government of India’s national language-technology mission.

The comparison is confined to *architectural properties*, and every cell is either a structural fact, a figure the vendor has published, or *n/d* where they have not. No row compares a measurement of ours against a number we could not obtain. Commercial products publish neither latency distributions nor retrieval quality nor memory footprints, and a table of plausible estimates for them would be a table of invented numbers set in the typography of measurement.

TABLE I

COMPARISON WITH DEPLOYED SYSTEMS. *n/d*: NOT DISCLOSED BY THE OPERATOR. ROWS ARE ARCHITECTURAL PROPERTIES, NOT PERFORMANCE CLAIMS.

Property	Khanmigo	BYJU’S	Bhashini	Ours
NCERT-aligned corpus	no	yes	—	yes
Indian languages	no	2	22	11
Corpus held locally	no	no	—	yes
Retrieval runs locally	no	no	—	yes
Generation runs locally	no	no	—	no
Runs without a network	no	no	no	no
Brahmic readability aid	no	no	no	yes
Per-answer grounding	n/d	n/d	—	yes
Corpus corruption audit	n/d	n/d	—	yes
Stated memory target	n/d	n/d	n/d	4 GB
Tutoring function	yes	yes	no	yes

Bhashini is dashed rather than marked absent on several rows because it is language infrastructure—translation, speech and transliteration services—and not a tutor, so corpus and retrieval questions do not apply to it. It is included because it is the most capable Indian-language system in public deployment, and because its 22 languages [22] exceed ours: a platform claiming multilingual reach should be compared against the strongest such system rather than only against tutors.

Three rows are stated against our own interest. Generation does not run locally, the system does not work without a network, and BYJU’S covers a curriculum we cover while operating at a scale we do not approach. What remains is the distinguishing set: the corpus, index and retrieval are local; a grounding score accompanies every answer; and the ingestion pipeline audits its own source material rather than embedding whatever it extracts. We have found no comparable system that reports the last two at all, which is a claim about the published record and not about what those systems do internally.

III. SYSTEM ARCHITECTURE

A. Request Lifecycle

A student’s question traverses an eight-stage pipeline from input to response. Table II summarizes the stages and measured latencies on an 8 GB Apple M1.

TABLE II
REQUEST LIFECYCLE LATENCY BENCHMARKS

Stage	Operation / Component	Latency
1	Rewrite to English / LLaMA-3.1-8B	~0.6 s
2	Embed query / BGE-M3 (fp16)	~0.1 s
3	Wide retrieval / pgvector HNSW	~0.02 s
4	Class scoring / Sum-of-top-3	<0.01 s
5	Narrow retrieval / pgvector HNSW	~0.02 s
6	Generate answer / LLaMA-3.1-8B	~2.0 s
7	Reading support (opt.)	~0.5 s
8	Illustration (opt.) / FLUX.1-dev	~5.0 s

Without optional stages, a warm request completes in approximately three seconds. The design separates the question’s language from the answer’s language. A student may type in romanized Hindi and receive a response in Tamil. This independence is critical because NCERT publishes in only three languages while the platform serves ten. Translation is a serving-layer responsibility handled by IndicTrans2 [7].

B. Model Stack

The platform integrates twenty-eight model identifiers, of which **five** are exercised on the critical path and measured in this work; the remainder are configured alternatives that this evaluation does not characterize. Table III separates the two, because a model stack presented as a single list would imply an empirical basis that only the first group has.

TABLE III

MODEL STACK. UPPER BLOCK: MEASURED ON THE CRITICAL PATH. LOWER BLOCK: INTEGRATED, NOT EVALUATED HERE. † DENOTES A HOSTED MODEL; ALL OTHERS ARE RESIDENT LOCALLY.

Role	Model	Params
Embeddings, grounding	BGE-M3	568M
Generation, rewriting	LLaMA-3.1-8B†	8B
Illustration	FLUX.1-dev†	12B
Vision OCR	Nemotron-Nano-12B†	12B
Generation (compared)	LLaMA-3.3-70B†	70B
Translation	IndicTrans2-1B	1B
Speech-to-Text	Whisper V3 Turbo [23]	809M
Text-to-Speech	MMS-TTS (11 ckpts)	~100M
Curriculum Valid.	Gemma-2-2B-IT	2B
Text Simplification	Qwen2.5-3B	3B
Reranking	BGE-Reranker-v2-M3	568M

Exactly one model is resident locally on the critical path. This is not an accident of packaging but the property that makes the memory budget of Section VI achievable: embedding must be local because every chunk in the corpus passes through it during ingestion, whereas generation, illustration and OCR are per-request and therefore hosted, leaving no language-model weights resident during serving.

BGE-M3 is used twice for distinct purposes—retrieving passages, and later scoring how far an answer strayed from

them (Section V-D). Both uses depend on the same property, cross-lingual embedding in a shared space, which is what permits a Devanagari passage and an English answer to be compared numerically at all.

Two integrated models are named but unusable in this deployment and are reported as such rather than omitted. The local OCR path (GOT-OCR2) requires CUDA and cannot execute on Apple silicon; its Python module additionally failed to import because `torchvision` was an undeclared dependency, which presented as a hardware limitation for some time before being diagnosed as a packaging one. The Tesseract fallback is installed without Devanagari training data, leaving it unable to read the script that most needs recovery (Section IV-C).

The choice of the 8B model over the 70B variant was driven by measurement (Table IV).

TABLE IV
TEXT GENERATION MODEL LATENCY COMPARISON

Model	Latency
meta/llama-3.1-8b-instruct	0.9 s
nvidia/nemotron-nano-9b-v2	2.5 s
meta/llama-3.3-70b-instruct	17–125 s
nvidia/llama-3.1-nemotron-nano-8b-v1	Timeout

C. Data Layer

The persistence layer combines PostgreSQL 17 with pgvector for HNSW-indexed similarity search, Redis 7 for response caching, and SQLite for the third cache tier. The multi-tier architecture (L1: in-memory LRU → L2: Redis → L3: SQLite) serves read-heavy endpoints. Cache keys incorporate a SHA-256 hash of the authorization header to prevent cross-user leakage. The vector index uses cosine distance:

```
CREATE INDEX idx_emb_hnsw_cosine
ON embeddings USING hnsw
(embedding vector_cosine_ops)
WITH (m=16, ef_construction=64);
```

An earlier migration stored embeddings as `double precision[]` rather than `vector(1024)`, making index creation impossible and condemning every search to a sequential scan.

D. Hardware Portability

The 4 GB target is a claim about the machines schools own, and those are predominantly Windows desktops rather than the Apple hardware this work was developed on. Portability is therefore part of the claim rather than an afterthought.

Device selection is resolved at load time in the order CUDA, Metal (MPS), CPU, and half precision is selected per device (Section VI-A). Everything the platform needs to know about its host—installed memory, available memory, swap, peak resident set—is read through a single abstraction over `psutil`, which reports identically on Linux, Windows and macOS.

That abstraction replaced direct system calls that answered only on macOS: `sysctl hw.memsize` for installed memory, `memory_pressure` and `sysctl vm.swapusage` for the ingestion memory guard, and `getrusage` for peak footprint. One call site was not inside an exception handler, so on Windows it raised rather than degrading; and the benchmark that substantiates the 4 GB figure imported the POSIX-only `resource` module at module scope, making it unrunnable on precisely the platform whose claim it exists to support.

Peak resident set is the one quantity with no portable source: `getrusage` is POSIX-only and Windows exposes a peak through the process handle. The abstraction reads whichever exists and returns a flag when it has had to fall back to current resident set, rather than presenting a smaller number as a high-water mark. A related unit bug is worth recording because it is silent: `ru_maxrss` is bytes on macOS and kilobytes on Linux, so the same code reports figures a factor of 1024 apart on two POSIX systems.

All measurements in this paper were taken on an Apple M1 with 8 GB of unified memory. We report no figures for hardware we did not run on.

IV. CORPUS CONSTRUCTION

A. The NCERT Catalog

NCERT encodes every textbook with a five-character code—`jesc1` represents Class 10, English medium, Science, book 1. The subject letters are not algorithmically derivable: science is `sc` for classes 7–10 but `cu` (Curiosity) for class 6, and splits into `ph`, `ch`, `bo` at classes 11–12. The catalog is scraped from NCERT’s textbook picker and cached locally.

The complete catalog contains **559 textbooks**: 209 in English, 191 in Hindi, and 159 in Urdu. The platform teaches from a deliberate subset of **263 books**—every English edition, plus the 54 subjects that exist only in Devanagari.

The 159 Urdu editions are excluded from the taught corpus entirely, and are neither ingested nor retrieved from. They are translations of books already held in English, so including them would triple the storage for that curriculum while adding no content, and cross-lingual retrieval was observed to cite an Urdu chapter as the source of an English answer—a citation the student can neither read nor check. Urdu remains available as an *answer* language, and appears as a query language in the retrieval evaluation of Section V-C: what a student writes in and reads back is independent of which edition the system learned from.

B. Ingestion Pipeline

The pipeline processes each book through five stages: ZIP download, per-chapter PDF extraction via PyMuPDF, text cleanup, semantic chunking (1200 characters, 200-character overlap, paragraph-boundary preference), BGE-M3 embedding (fp16), and storage in PostgreSQL. Each book is committed as a single database transaction for resumability. Download and embedding contend for no shared resource: one waits on a remote web server, the other saturates the GPU. Executed sequentially they idle in turn, so the next book’s archive

is fetched on a prefetch thread while the current book is embedded.

The speedup was established by a controlled comparison rather than inferred. The same three books (classes 6–8 *Curiosity*) were ingested twice, once with prefetching disabled through `INGEST_PREFETCH=0`, producing byte-identical output in both runs (37 chapters, 1,130 chunks): **193 s per book sequentially against 140 s per book with prefetching, a 1.38 $\times$ speedup**. Extrapolated, the 153 obtainable books require approximately 6.0 hours on a single Apple M1. The figure is quoted over the obtainable books rather than the full 263-book curriculum, because the remainder cannot be ingested at any rate.

We note that a naive comparison of figures taken under different conditions would have reported this backwards. An earlier sequential measurement of 97 s per book was recorded on a different, smaller book set; read against the 140 s figure it appears to show prefetching making ingestion *slower*. Only the paired measurement on identical inputs is informative, and unpaired throughput figures are excluded here for that reason.

C. Legacy Font Corruption

Two-thirds of the Hindi curriculum is rendered unreadable by every PDF text extractor tested. NCERT’s older Hindi textbooks are typeset in **Walkman-Chanakya 905**, a legacy font mapping Devanagari glyphs onto ASCII codepoints with no `ToUnicode` table. Text extraction returns raw byte values. The analysis covers 41 of the 54 Devanagari-only books in the taught curriculum: the method samples a single chapter PDF per book, and for the remaining 13 books NCERT publishes no individual chapter files, returning HTTP 404 for chapters 1 through 3 at the documented URL scheme. Those 13 are absent for lack of a retrievable sample rather than by selection, and no result here is conditioned on their contents (Table V):

TABLE V
HINDI CORPUS CORRUPTION ANALYSIS

Extraction Category	Books	Devanagari
Legacy font (Walkman-Chanakya)	27	0.0%
Unicode font (Kokila, Arya)	14	97.9–100%

The measurement that was inverted. An earlier quality assessment counted broken Devanagari sequences and reported the 27 legacy-font books as the *cleanest* in the corpus—a detector searching for damaged Devanagari finds nothing in a book containing no Devanagari at all. 133 chunks of Latin gibberish entered the retrieval corpus before the error was caught. The corrected detection identifies Hindi books whose extracted text is overwhelmingly Latin.

D. Text Layer Fidelity in Mathematics

PyMuPDF’s text layer is lossy on mathematical content. Class 10 Mathematics chapter 4 defines a quadratic as $ax^2 + bx + c$, $a \neq 0$; the text layer yields “ $ax2 + bx + c$, $a \ 0$ ”—the superscript flattened and the inequality deleted. Pages requiring OCR are identified by image density (Table VI).

TABLE VI
IMAGE DENSITY AS OCR PREDICTOR

Page Type	Images/1000 chars
Mathematics with equations	30.8
Science with figures	5.9
Prose with occasional figures	2.0–4.0

Cloud OCR (Nemotron VL) correctly recovers symbols but introduces substitution errors worse for learners: the text layer loses glyphs; the vision model substitutes plausible but incorrect characters and hallucinates nonexistent headers.

V. CROSS-LINGUAL RETRIEVAL

A. Evaluation Corpus

Every retrieval and grounding result in this section and the next was measured against a single corpus state, recorded here so that the numbers are interpretable and reproducible. The database held **153 textbooks**—138 English and 15 Hindi—comprising 1,542 chapters and **42,995 indexed passages**, with an embedding present for every passage (42,995 of 42,995; a passage lacking one is invisible to retrieval and is not otherwise detectable). All twelve classes are represented, ranging from 3 books at class 1 to 34 at class 12.

This is 153 of the 263-book taught curriculum. The remaining 110 are not pending but unobtainable: NCERT publishes no archive for most of them, one class 5 archive is corrupt, and the rest are the legacy-font Hindi readers of Section IV-C, which the pipeline rejects rather than store as gibberish. Each is recorded with its reason, so the shortfall is a property of the source material rather than of ingestion time.

Composition is load-bearing for one result rather than incidental to it. English science and mathematics books outnumber the Hindi readers roughly nine to one, which is what makes the grade-attribution result of Section V-Da test rather than a formality: a retriever that defaults to the majority of its corpus would answer every Hindi-literature question from a science chapter.

Absolute cosine similarities should be read as characteristic of this corpus and not as a property of the method. Recall in particular is not estimated: there is no ground-truth relevant set for these books, and no such claim is made.

B. The Romanized Hindi Problem

BGE-M3’s cross-lingual capability does not extend to romanized Hindi—the predominant input mode for Indian students using standard keyboards. The same chemistry question was embedded in three scripts (Table VII).

TABLE VII
SCRIPT IMPACT ON RETRIEVAL QUALITY

Script	Top Retrieval	Correct
English	Class 10 at 0.607	Yes
Devanagari	Class 10 at 0.564	Yes
Romanized Hindi	Class 1 at 0.520	No

Romanized Hindi drifts to class 1 picture books matching on surface-level token overlap. This failure provides a natural ablation study comparing naive direct retrieval (MultiRAG) against a translation-then-retrieval (tRAG) approach. Direct retrieval across scripts failed entirely. By rewriting every question into a short English search query before embedding (following Ma et al. [4]), the vector search is successfully forced back into the correct semantic neighborhood.

C. Cross-Lingual Retrieval Quality

Retrieval was evaluated over **22 queries: two topics in each of the eleven supported Indian languages**, against the English-medium portion of the corpus. Holding the topic constant across languages makes language the only variable that moves. One topic uses ordinary vocabulary (the focusing of the human eye); the other is *photosynthesis*, whose name in most of these languages is a compound of morphemes meaning “light” and “joining”. The queries are committed with the benchmark rather than only their results.

Five configurations were measured (Table VIII), from a lexical baseline to the deployed system. A query counts as correct only when the *top* passage is on topic.

TABLE VIII
CROSS-LINGUAL RETRIEVAL: 22 QUERIES OVER 11 LANGUAGES, AGAINST THE
ENGLISH-MEDIUM CORPUS. CORRECT = ON-TOPIC PASSAGE AT RANK 1.

Configuration	Eye	Photo.	Overall
BM25 (lexical)	0/11	0/11	0/22 (0%)
BM25 + rewrite	6/11	4/11	10/22 (45%)
Dense (BGE-M3)	9/11	2/11	11/22 (50%)
Dense + rewrite	6/11	8/11	14/22 (64%)
Pooled (deployed)	8/11	8/11	16/22 (73%)

The lexical baseline retrieves nothing at all. BM25 returns zero hits for all 22 queries, not merely poor ones: a Devanagari, Tamil or Perso-Arabic query shares no tokens with an English corpus, so the inverted index has nothing to match. Lexical retrieval is not weaker at this task, it is structurally incapable of it, and that is the argument for carrying a 568 M-parameter embedding model on a 4 GB machine. Given an English query from the rewrite it becomes viable at once (45%), which locates the difficulty precisely: the barrier is the script, not the vocabulary.

The two arms fail on different inputs, and that is the result. Embedding the question directly handles ordinary vocabulary well (9 of 11) and collapses on compound terms (2 of 11), because the embedding attends to the constituent morphemes and retrieves optics for a word that means “light-joining”. Rewriting inverts both: it normalises the compound to its English canonical form and recovers 9 of 11, while losing three ordinary-vocabulary cases whose original phrasing had been more precise than its English paraphrase.

Neither is therefore the right configuration on its own. Pooling both retrievals scores **16 of 22**, above either arm, because the union recovers what each misses rather than averaging them.

The rewrite is a language-model call and is not deterministic at temperature zero: repeated runs of this benchmark returned

16 and 17 of 22, so the pooled figure should be read as 16 ± 1 . The ordering of the five configurations was stable across runs, which is the claim the table is making. This is an empirical justification for a design that had been adopted on reasoning alone, and it also explains the earlier failure mode in which a mistranslated rewrite (“Rahim” rendered as “Rahman”) silently replaced a question that would have retrieved correctly.

The one query that fails in every configuration is Urdu photosynthesis, whose term (*ziyai talif*, Perso-Arabic in origin) shares no morphology with either the English word or the Sanskrit-derived compounds the other ten languages use—so neither arm has a handle on it.

D. Grade-Level Classification

The platform uses the **sum of each class’s three strongest passage similarities** as the classification signal. Two earlier functions were rejected: summing all similarities allowed volume to dominate (eight weak class 1 matches outscored three strong class 10 matches); averaging exhibited the opposite failure (a single high-similarity passage won by being its own mean). The `vote_grade` function lets each query independently propose a class on its own scale, weighted by decisiveness.

VI. MEMORY OPTIMIZATION

A. Half-Precision Embeddings

BGE-M3’s weights were evaluated at both full (float32) and half (float16) precision (Table IX).

TABLE IX
PRECISION COMPARISON FOR BGE-M3

Metric	float32	float16
Weight memory	2,166 MB	1,083 MB
Encode 4 queries	2,255 ms	1,197 ms
Model load time	8.5 s	8.1 s
Cosine fidelity	—	0.999999

Half precision yields 50% memory reduction and $\sim 1.88\times$ speedup with no measurable quality degradation. A cosine similarity of 0.999999 cannot reorder any result list.

B. Serving Footprint: Measurement vs. Arithmetic

An earlier analysis summed component sizes to ~ 1.6 GB and concluded comfortable fit within 4 GB. A dedicated benchmarking script measures actual peak RSS through `getrusage(RUSAGE_SELF)`—a high-water mark surviving page eviction, unlike `ps` which reported 46 MB for a process holding a 1 GB model (Table X).

The actual peak is **2,094 MB**, not the 1,600 MB predicted. Loading model weights accounts for only 534 MB (`safetensors` memory-maps them); the first forward pass adds 1,560 MB as weight pages become resident and attention workspaces are allocated. The system fits within 4 GB with ~ 1.4 GB usable headroom (accounting for ~ 500 MB of operating system overhead), not the 2.4 GB arithmetic implied.

TABLE X
SERVING PIPELINE MEMORY FOOTPRINT (MEASURED)

Pipeline Stage	Peak RSS
Interpreter start	14 MB
Configuration imported	265 MB
BGE-M3 loaded (fp16)	534 MB
Query encoded	2,094 MB
pgvector search complete	2,094 MB

C. Ingestion Memory Management

A single process ingesting 263 books reached book 8 and entered an unrecoverable state: swap at 11.4 GB of 12.2 GB maximum, the process in uninterruptible I/O wait. `SIGKILL` is not delivered in that state. The solution is batch processing: one process per six books, with ~ 20 seconds of model reloading per batch.

VII. READING SUPPORT FOR BRAHMIC SCRIPT READERS

A. Akshara Segmentation

Dyslexia support tooling is overwhelmingly designed for English. Most Indian students read a Brahmic script where the fundamental unit is the **akshara**—a consonant cluster carrying an inherent or explicit vowel. The segmentation algorithm recognizes the virama (halant) in each Brahmic script as the signal that two consonants are conjoined, operating generically across Devanagari, Bengali, Gurmukhi, Gujarati, Odia, Tamil, Telugu, Kannada, and Malayalam.

Two details: (1) sentence counting must honor the danda, not only the Latin period; (2) Flesch-Kincaid is computed for English only—akshara counts are not syllable counts.

B. Measured Simplification Quality

The first simplification attempt produced text *harder* than the original (Table XI).

TABLE XI
READABILITY MEASUREMENT: FLESCH-KINCAID GRADE LEVEL

Version	FK Grade
NCERT source text	7.0
First simplification attempt	8.2 (harder)
Measure-and-keep approach	6.9 (easier)

The corrected approach measures the original, rewrites with readability as sole objective, and retains whichever version is easier.

C. Evidence-Based Typography

Letter and word spacing (on by default) is the only intervention with strong evidence [13]. Dyslexia-specific fonts are offered as preferences but labeled as weakly evidenced [14], [15]. Nothing in the module diagnoses any condition; it adjusts presentation.

TABLE XII
SECURITY AUDIT FINDINGS (SELECTED)

#	Vulnerability / Consequence
1	Safety pipeline import error caught by bare <code>except</code> : every response marked safe without evaluation
2	JWT type claim unchecked: 7-day refresh tokens accepted as 30-minute access tokens
3	<code>validate_required()</code> never called: production booted with no JWT secret
5	Embedding column as double <code>precision[]</code> : no vector index possible
7	RLS compared <code>NULL=NULL</code> : shared curriculum invisible to all users
9	Sentry received student PII despite <code>send_default_pii=False</code>

VIII. SECURITY ANALYSIS

A comprehensive audit identified twelve vulnerabilities, all repaired (Table XII).

Vulnerability #1 is the most consequential: the safety pipeline was designed as a three-pass content filter, but an import path error caused the entire pipeline to fail to load. The error was caught by a bare `except` clause returning `(response, True)`, making the content safety system structurally incapable of blocking any response since deployment.

IX. GROUNDING AND HALLUCINATION DETECTION

A retrieval-augmented system can retrieve the correct chapter and still generate a fabricated answer. When asked about a specific story from the Hindi curriculum, the system retrieved the correct chapter, then described events that never occurred. The class was correct, the chapter was correct, the language was correct, and every detail was invented.

The **grounding score** computes cosine similarity between the generated answer and the source passages. Lexical overlap cannot serve this purpose because the platform produces answers in a different language from the source passages; BGE-M3’s cross-lingual embedding property makes the metric functional in exactly this case.

A. Calibration on a Distribution

The threshold was originally set at 0.55 from four hand-verified answers, three correct at 0.583–0.701 and one fabricated at 0.489. Four points that happen to separate are not a calibration, and the difficulty is obtaining ungrounded answers in quantity: a model instructed to hallucinate does not fail the way a grounded pipeline fails.

The negatives are therefore constructed rather than solicited. Every answer is generated normally from its own retrieved passages, then scored twice: against the passages it was written from (*matched*), and against another question’s passages (*mismatched*). A mismatched pair is precisely the failure the metric exists to catch—fluent, well-formed, on a different subject—and it requires no human labelling and no second generation pass. Sixteen questions spanning subjects and classes yield 16 matched and 240 mismatched pairs.

Grounding need not be measured against the concatenation of the retrieved passages, so four estimators were compared over identical generations (Table XIII). The two runs of the experiment, with independently generated answers, agreed to within 0.007 on every entry.

TABLE XIII

GROUNDING ESTIMATORS OVER 16 MATCHED AND 240 MISMATCHED PAIRS. BALANCED ACCURACY IS AT EACH ESTIMATOR’S OWN BEST THRESHOLD; CHANCE IS 0.500.

Estimator	Match	Mismatch	Gap	t	Acc.
Concatenated	0.830	0.470	0.360	0.715	1.000
Best passage	0.854	0.502	0.352	0.720	1.000
Mean of top 3	0.822	0.486	0.336	0.695	0.996
Per sentence	0.676	0.455	0.221	0.555	0.985

The simplest estimator is also the best: scoring the answer against all retrieved passages joined separates the two populations by 0.360, and no refinement improves on it. Scoring against the single best passage raises both means without widening the gap, and scoring sentence by sentence narrows it—a correct answer contains connective and pedagogical sentences that appear in no passage, and averaging over them dilutes the signal the metric depends on.

B. The Threshold Was Too Low

The distribution shows that 0.55 separates the populations but sits in the wrong place. Thirty-five of the 240 mismatched pairs—one drifted answer in seven—score above it and pass as grounded. At 0.715 none do, while no matched answer falls below it: balanced accuracy rises from 0.927 to 1.000, and the number of correct answers needlessly regenerated stays at zero. The deployed threshold was raised accordingly. The original four points were not wrong, merely too few to locate the boundary.

X. CONSOLIDATED RESULTS

A. Validation Suite

The platform is validated by **445 automated tests** with zero failures: ~ 350 unit, ~ 60 integration, ~ 20 end-to-end, and ~ 15 performance benchmark tests.

B. System-Level Metrics

Table XIV summarizes key metrics.

XI. DISCUSSION

The measurement-driven methodology reveals a recurring pattern: intuitive engineering assumptions are frequently contradicted by empirical measurement, and the contradictions are more informative than the assumptions.

The legacy font detection illustrates this most starkly. A quality metric searching for damaged Devanagari ranked the most corrupted books as the best—because corruption manifested as the *absence* of Devanagari, which the metric interpreted as perfection. The memory estimation follows the same pattern: summing static sizes to 1.6 GB misses the

TABLE XIV
CONSOLIDATED SYSTEM METRICS

Metric	Value
NCERT catalog coverage	559 textbooks
Taught curriculum	263 textbooks
Ingested corpus (all results)	153 textbooks
Supported languages	11
API routes	72
Ingestion throughput	140 s/book ($1.38\times$)
Ingestion time, 153 books	~ 6.0 hours
Peak serving memory (fp16)	2,094 MB
fp16 cosine fidelity	0.999999
Memory reduction (fp16)	50%
Encoding speedup (fp16)	$1.88\times$
Response latency (warm)	~ 3 s
Grounding threshold (calibrated)	0.715
Grounding balanced accuracy	1.000
Security vulns fixed	12
Automated tests passing	445/445

2.1 GB transient peak. The grounding score addresses a fundamental RAG limitation: a system can retrieve correctly, generate fluently, pass every surface check, and still fabricate content.

Limitations. (1) The grounding negatives are constructed by mismatching answers to other questions’ passages rather than collected from observed hallucinations; this captures topical drift, which is the failure we have seen, but not an answer that stays on topic and invents a detail within it. (2) Cross-lingual retrieval is evaluated on two topics per language; broader topic coverage would tighten the estimate. (3) 27 legacy-font Hindi books remain unprocessable. (4) Memory measurements have run-to-run variance of up to 414 MB.

XII. CONCLUSION AND FUTURE WORK

This paper presented Shiksha Setu, a local-retrieval AI tutoring platform whose pipeline addresses the complete NCERT catalog of 559 textbooks and which delivers instruction in eleven Indian languages on consumer hardware with as little as 4 GB of RAM. The reported evaluation rests on 153 ingested textbooks and 42,995 indexed passages spanning all twelve classes. Every performance claim has been substantiated through kernel-level measurement, and engineering failures have been documented alongside their corrections.

The principal contributions are: (1) corpus construction with legacy-font corruption detection affecting 66% of Hindi textbooks, (2) cross-lingual retrieval measured over 22 queries in eleven languages, with an ablation showing that pooling the raw and rewritten queries retrieves correctly for 16 ± 1 of 22 against 11 and 14 for either alone, and that a lexical baseline retrieves none, (3) a grounding metric separating grounded from drifted answers by 0.360 in cosine similarity over 256 pairs, at a calibrated threshold that admits none of the 240 negatives where the previously shipped value admitted 35, (4) half-precision inference halving memory at 0.999999 cosine fidelity, (5) akshara-level readability for nine Brahmic scripts, and (6) a security audit repairing twelve production vulnerabilities.

Future work targets: conducting a human-in-the-loop pilot study with actual educators to evaluate pedagogical utility,

expanding the hallucination annotation dataset (100+ response pairs) for robust grounding calibration, integrating a cross-encoder reranker to resolve compound-term failures (e.g., photosynthesis), implementing Entity Linking for polysemous term disambiguation, and developing a Walkman-Chanakya-to-Unicode transliterator.

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